

14 The equation for a reaction is shown.



Substances P, Q, R and S are either all in solution or are all gases.

The equilibrium concentrations of P, Q, R and S are measured in three experiments. The experiments have different starting concentrations and are performed under different conditions. The results are shown in the table.

	experiment 1	experiment 2	experiment 3
pressure	100 kPa	100 kPa	200 kPa
temperature	300 K	400 K	300 K
[P] at equilibrium	0.050 mol dm ⁻³	0.040 mol dm ⁻³	0.080 mol dm ⁻³
[Q] at equilibrium	0.064 mol dm ⁻³	0.042 mol dm ⁻³	0.032 mol dm ⁻³
[R] at equilibrium	0.075 mol dm ⁻³	0.070 mol dm ⁻³	0.015 mol dm ⁻³
[S] at equilibrium	0.040 mol dm ⁻³	0.035 mol dm ⁻³	0.080 mol dm ⁻³

Which conclusion can be drawn from these results?

- A The reactants are all gaseous.
- B The reactants are all in solution.
- C The forward reaction is endothermic.
- D The forward reaction is exothermic.